

Highlights

Initial-Trace Insufficiency and Entropy-Accounting Separation in Finite-Volume Coarse Graining

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- Initial traces close forward Euler but can fail for Heun updates.
- Realized stage-integrated fluxes close completed finite-volume transitions.
- Conservative registers do not determine entropy-accounting information.
- A Burgers–Rusanov sign family gives exact entropy-blindness error floors.

Initial-Trace Insufficiency and Entropy-Accounting Separation in Finite-Volume Coarse Graining

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ABSTRACT

Coarse finite-volume information does not automatically define an exact pathwise closure for the restricted evolution of a fine-grid realization. We study two one-step information boundaries that are distinct from recursive predictive memory. First, we show that parent averages together with complete initial parent-interface trace pairs are sufficient for one forward-Euler finite-volume transition with a deterministic two-state numerical flux, but they do not generally determine a completed second-order strong-stability-preserving Runge-Kutta, equivalently Heun, transition. An explicit scalar upwind collision shows that hidden interior data can affect later stage traces and hence the completed parent update, even when initial traces and parent averages agree.

Second, we show that exact conservative one-step closure is not the same as entropy-accounting closure. A parent entropy balance obtained from a cellwise numerical entropy balance contains a subcell Jensen gap, a realized entropy-flux difference, and residual dissipation. For a declared Burgers-Rusanov conservative flux and a declared Rusanov-type numerical entropy flux, we construct a sign-indexed finite family in which all branches have identical parent averages, identical conservative interface registers, and identical next parent averages, while the entropy-interface register distinguishes every branch. Under the uniform branch law, the conservative channel leaves nonzero deterministic squared-error floors for both the raw entropy register and its cyclic divergence. The results give a concise finite-volume information-sufficiency test for multistage interface data and entropy-side accounting.

1. Introduction

Finite-volume methods evolve cell averages through numerical fluxes at cell interfaces [1]. In coarse/fine coupling, adaptive mesh refinement, and reduced modelling, one often asks whether a declared coarse information channel is sufficient to reproduce the restricted result of a fine-grid computation [2–6]. Parent averages alone generally do not determine subcell arrangement, and subcell arrangement can affect later interface fluxes.

A companion paper studies the recursive version of this issue for a periodic scalar upwind hierarchy: it determines exact finite-horizon memory, conditioning, and dissipative decay for future parent-average prediction [7]. The present paper asks a different one-step question: which declared interface and entropy-side data close a completed finite-volume transition, and which proposed channels do not?

The one-step conservative baseline is classical. If a completed fine finite-volume step has shared conservative face fluxes, then summing child cell balances over a parent cancels internal fluxes and leaves only the parent-interface flux difference. This is the algebra behind refluxing and conservative mesh-interface communication [2, 3]. The realized integrated interface flux is therefore a canonical ex post sufficient message for reconstructing the restricted result of an already completed fine step. It is not, by itself, a cheap autonomous predictor before the fine work is done.

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The new points here concern two boundaries of this one-step baseline. The first is a multistage boundary. For forward Euler, complete ordered initial traces at a parent interface determine the parent-interface numerical flux. For a multistage method, later stage traces can depend on hidden interior fine data. We give an exact same-data collision for the second-order strong-stability-preserving Runge–Kutta method, denoted SSPRK(2,2), also known as the explicit trapezoidal or Heun method in this linear autonomous setting.

The second boundary is an entropy-accounting boundary. Entropy-stable and entropy-accounting finite-volume methods require information beyond the conservative update [8–12]. A parent entropy balance involves a subcell Jensen gap, a realized entropy-flux difference, and residual dissipation. We construct an exact Burgers–Rusanov family where the conservative channel is identical across branches while the entropy-side register is not.

These questions are also relevant to data-driven discretizations and closure models [13–16]. Exact unresolved finite-volume terms can be written down for declared filters and discretizations [17]; the present paper instead asks whether a proposed retained channel is sufficient for a target transition or entropy account to be single valued.

Contributions. The factorization criterion and flux telescoping identity used below are classical. The contributions of this paper are:

1. a one-step information-channel formulation for finite-volume coarse-graining contracts;
2. a proof that initial parent-interface traces close forward Euler but can fail for SSPRK(2,2)/Heun, even when parent averages also agree;
3. an exact Burgers–Rusanov family showing that conservative one-step closure does not determine the augmented parent entropy account.

Organization. Section 2 states the factorization and flux-register baselines in minimal form. Section 3 proves the initial-trace sufficiency and SSPRK(2,2)/Heun insufficiency results. Section 4 derives the parent entropy account and the entropy-blind finite family. Section 5 discusses implications and limitations, and Section 6 concludes.

2. One-Step Closure Baseline

Let $\mathcal{S}$ be a set of admissible fine states, let

$$T : \mathcal{S} \rightarrow \mathcal{C}$$

be a declared coarse information map, and let

$$\mathcal{H} : \mathcal{S} \rightarrow \mathcal{Y}$$

be the target restricted output.

Lemma 2.1 (Fiber factorization). *There exists a unique map $G : T(\mathcal{S}) \rightarrow \mathcal{Y}$ such that*

$$\mathcal{H} = G \circ T$$

if and only if

$$T(x) = T(x') \implies \mathcal{H}(x) = \mathcal{H}(x') \quad (x, x' \in \mathcal{S}).$$

Proof. If $\mathcal{H} = G \circ T$, equal T -values imply equal $\mathcal{H}$ -values. Conversely, if $\mathcal{H}$ is constant on fibers of T , define $G(t)$ as the common value of $\mathcal{H}$ on $T^{-1}(t)$. This is well defined on $T(\mathcal{S})$ and is unique there. $\square$

In a one-dimensional finite-volume hierarchy, a completed conservative fine step has the form

$$U_i^{n+1} = U_i^n - \frac{\Phi_{i+1/2} - \Phi_{i-1/2}}{h_i},$$

where $\Phi_{i+1/2}$ is the realized time-integrated fine-face flux. If parent K contains the fine cells $i \subset K$, and

$$H_K = \sum_{i \subset K} h_i, \quad (RU)_K = \frac{1}{H_K} \sum_{i \subset K} h_i U_i,$$

then summing the child balances gives

$$(RU^{n+1})_K = (RU^n)_K - \frac{Q_K - Q_{K-1}}{H_K}, \quad (1)$$

where Q_K is the realized integrated flux through the parent interface $K + 1/2$. On a periodic parent grid, only the cyclic differences $Q_K - Q_{K-1}$ affect the parent update; adding one spatially constant flux register changes no parent average. This elementary gauge is the one-step counterpart of the periodic gauge used in recursive predictive-memory analysis [7].

Equation (1) is used here only as the classical baseline: the realized integrated flux register is sufficient for the result of a completed step. The next section asks when a simpler initial-trace channel can replace that register.

3. Initial Traces and Multistage Flux Registers

Throughout this section the mesh, time step, flux parameters, and parent partition are fixed, there are no source or nonconservative terms, and all boundary fluxes are either periodic or externally supplied. A trace at a parent interface means the actual ordered one-sided state pair passed to the deterministic numerical flux.

Proposition 3.1 (Forward-Euler trace sufficiency). *Consider a one-dimensional finite-volume forward-Euler step*

$$U_i^{n+1} = U_i^n - \frac{\Delta t}{h_i} \left[\hat{F}(U_i^n, U_{i+1}^n) - \hat{F}(U_{i-1}^n, U_i^n) \right],$$

where $\hat{F}$ is a deterministic two-state numerical flux with known parameters. If the declared information consists of the current parent averages and the complete ordered initial trace pair at every parent interface, then the next parent averages are determined exactly.

Proof. Summing the fine updates over a parent cancels all internal fine-face fluxes. The only remaining terms are the two parent-interface numerical fluxes

$$\hat{F}(U_{K,r-1}^n, U_{K+1,0}^n) \quad \text{and} \quad \hat{F}(U_{K-1,r-1}^n, U_{K,0}^n).$$

These are determined by the ordered initial traces. The corresponding integrated forward-Euler registers are Δt times these numerical fluxes. Together with the current parent averages, they determine the next parent averages through (1). $\square$

This forward-Euler sufficiency is special to a one-stage update. In a multistage method, later stage traces may depend on subcell values that are not visible in the initial parent-interface traces.

Let A_λ be the periodic scalar positive-upwind forward-Euler matrix,

$$(A_\lambda u)_i = (1 - \lambda)u_i + \lambda u_{i-1}, \quad 0 < \lambda \leq 1,$$

with periodic indexing. For this linear autonomous update, the SSPRK(2,2), equivalently explicit trapezoidal or Heun, completed-step operator is

$$M_\lambda = I + (A_\lambda - I) + \frac{1}{2}(A_\lambda - I)^2 = \frac{1}{2}(I + A_\lambda^2).$$

Theorem 3.2 (Initial traces do not close SSPRK(2,2)/Heun). *Consider $P = 2$ parents, $r = 4$ fine cells per parent, and periodic scalar positive-speed upwinding. Let*

$$x^+ = (0, -1, 1, 0 \mid 0, 0, 0, 0), \quad x^- = (0, 1, -1, 0 \mid 0, 0, 0, 0),$$

where the vertical bar separates the two parents. The two states have identical parent averages and identical initial parent-interface traces. However,

$$RM_\lambda x^+ = \left(-\frac{\lambda^2}{8}, \frac{\lambda^2}{8}\right), \quad RM_\lambda x^- = \left(\frac{\lambda^2}{8}, -\frac{\lambda^2}{8}\right).$$

Thus parent averages and initial parent-interface traces do not determine the completed SSPRK(2,2)/Heun parent update.

Proof. Both states have zero parent averages, and the ordered trace pairs at both parent interfaces are $(0, 0)$ in both states. Let

$$v = x^+ - x^- = (0, -2, 2, 0 \mid 0, 0, 0, 0).$$

Then $Rv = 0$. If S denotes the periodic shift $(Su)_i = u_{i-1}$, then

$$A_\lambda^2 = (1 - \lambda)^2 I + 2\lambda(1 - \lambda)S + \lambda^2 S^2.$$

Only the S^2 term moves part of v across the parent interface. Direct summation over the two parents gives

$$RM_\lambda v = \left(-\frac{\lambda^2}{4}, \frac{\lambda^2}{4}\right).$$

Since $x^+ = v/2$ and $x^- = -v/2$, the displayed values follow. $\square$

The obstruction is not that stage-integrated fluxes fail. In a standard Butcher-form finite-volume Runge-Kutta step, the corresponding sufficient ex post register at parent interface $K + 1/2$ is

$$Q_K^{\text{RK}} = \Delta t \sum_s b_s \hat{F} \left(U_{K,r-1}^{(s)}, U_{K+1,0}^{(s)} \right),$$

with the declared stage values $U^{(s)}$ and weights b_s . These realized stage-integrated interface fluxes still reconstruct the restricted result of the completed fine step. The point is narrower and sharper: the initial trace channel that suffices for forward Euler does not factor the completed multistage transition.

4. Entropy-Accounting Separation

Conservative one-step closure determines next parent averages. It does not determine the augmented parent-level entropy account. Once the next parent mean is known, the entropy of that parent mean is known, but the decomposition into subcell Jensen gap, entropy-flux divergence, and residual dissipation is additional information.

Let η be a convex entropy on a convex state domain. Define the parent mean and subcell Jensen gap in parent K at time level m by

$$\bar{U}_K^m = (RU^m)_K, \quad \delta_K^m = \sum_{i \subset K} \frac{h_i}{H_K} \eta(U_i^m) - \eta(\bar{U}_K^m).$$

By Jensen's inequality, $\delta_K^m \geq 0$.

Theorem 4.1 (Augmented parent entropy balance). *Suppose the fine cells satisfy the exact entropy-amount balance*

$$h_i \eta(U_i^{n+1}) + \mathcal{D}_i = h_i \eta(U_i^n) - (\Psi_{i+1/2} - \Psi_{i-1/2}), \quad \mathcal{D}_i \geq 0,$$

for a declared numerical entropy flux $\Psi_{i+1/2}$. Let $\Psi_{K,+}$ and $\Psi_{K,-}$ denote the realized integrated entropy fluxes through the right and left parent interfaces. Then

$$\eta(\bar{U}_K^{n+1}) + \delta_K^{n+1} + \frac{\mathcal{D}_K}{H_K} = \eta(\bar{U}_K^n) + \delta_K^n - \frac{\Psi_{K,+} - \Psi_{K,-}}{H_K},$$

where

$$\mathcal{D}_K = \sum_{i \subset K} \mathcal{D}_i.$$

Proof. Sum the fine entropy balances over all children in parent K . Internal entropy fluxes telescope. The identity

$$\sum_{i \subset K} h_i \eta(U_i^m) = H_K [\eta(\bar{U}_K^m) + \delta_K^m]$$

then gives the displayed parent balance after division by H_K . $\square$

This theorem is conditional. It assumes a cellwise entropy balance with a declared numerical entropy flux and nonnegative residuals. It does not claim that every finite-volume or Runge–Kutta method automatically has such a balance. Different entropy-flux conventions can redistribute entropy flux and residual dissipation, so the information separation below is for the declared entropy pair and declared numerical entropy flux.

4.1. A finite entropy-blind Burgers–Rusanov family

Consider inviscid Burgers' equation

$$u_t + \left(\frac{u^2}{2} \right)_x = 0$$

with quadratic entropy and entropy flux

$$\eta(u) = \frac{1}{2}u^2, \quad q(u) = \frac{1}{3}u^3.$$

Use forward Euler with $\Delta t = 1/2$, unit fine cells, and the Rusanov conservative numerical flux with $\alpha = 1$,

$$\widehat{f}(u_L, u_R) = \frac{1}{2}(f(u_L) + f(u_R)) - \frac{1}{2}(u_R - u_L).$$

We pair it with the declared Rusanov-type numerical entropy flux

$$\widehat{q}(u_L, u_R) = \frac{1}{2}(q(u_L) + q(u_R)) - \frac{1}{2}(\eta(u_R) - \eta(u_L)).$$

The corresponding integrated conservative and entropy fluxes are

$$\Phi_{i+1/2} = \Delta t \widehat{f}(u_i, u_{i+1}), \quad \Psi_{i+1/2} = \Delta t \widehat{q}(u_i, u_{i+1}).$$

Lemma 4.2 (Local entropy residuals). *Let $a, b \in \{-1, +1\}$, and consider the four-cell local state*

$$(a, -a, -b, b),$$

with compatible exterior states a on the left and b on the right. After one Rusanov forward-Euler step with $\Delta t = 1/2$, the four updated values are

$$\left(\frac{a}{2}, -\frac{a+b}{4}, -\frac{a+b}{4}, \frac{b}{2} \right).$$

The residuals

$$\mathcal{D}_i = \eta(u_i^n) - \eta(u_i^{n+1}) - (\Psi_{i+1/2} - \Psi_{i-1/2})$$

are nonnegative and are listed in Table 1.

Table 1

Local entropy residuals for the four-cell Burgers–Rusanov stencil $(a, -a, -b, b)$, with $a, b \in \{-1, +1\}$. All entries are nonnegative, which verifies the cellwise entropy-balance hypothesis used in Theorem 4.1 for every local branch of the sign family.

(a, b)	$\mathcal{D}_0$	$\mathcal{D}_1$	$\mathcal{D}_2$	$\mathcal{D}_3$
$(1, 1)$	13/24	13/24	5/24	5/24
$(1, -1)$	13/24	1/2	1/2	13/24
$(-1, 1)$	5/24	1/2	1/2	5/24
$(-1, -1)$	5/24	5/24	13/24	13/24

Proof. For $u_L, u_R \in \{-1, +1\}$,

$$\widehat{f}(u_L, u_R) = \frac{1}{2} + \frac{1}{2}(u_L - u_R).$$

Substitution into the local forward-Euler update gives the displayed updated values. The numerical entropy flux satisfies

$$\widehat{q}(u_L, u_R) = \frac{u_L^3 + u_R^3}{6} - \frac{u_R^2 - u_L^2}{4}.$$

For sign states, $u_L^2 = u_R^2 = 1$, so the second term vanishes and

$$\Psi_{i+1/2} = \frac{u_i + u_{i+1}}{12}.$$

Substitution into the definition of $\mathcal{D}_i$ gives the four rows in Table 1. □

Theorem 4.3 (Conservative blindness to entropy-accounting data). *For every $P \geq 2$, let*

$$\sigma = (\sigma_0, \dots, \sigma_{P-1}) \in \{-1, +1\}^P.$$

In parent K , set the four child values to

$$(\sigma_{K-1}, -\sigma_{K-1}, -\sigma_K, \sigma_K),$$

with periodic indexing. Then all 2^P branches have identical parent means, identical conservative parent-interface flux registers, and identical next parent means. However, their raw entropy-interface registers are

$$\Psi_K(\sigma) = \frac{\sigma_K}{6},$$

which distinguish all branches. The entropy-flux part of the periodic parent entropy balance depends on the cyclic divergence

$$\beta_K(\sigma) = \Psi_K(\sigma) - \Psi_{K-1}(\sigma) = \frac{\sigma_K - \sigma_{K-1}}{6},$$

which takes $2^P - 1$ distinct values. Under the uniform branch law,

$$\mathbb{P}(\beta = 0) = 2^{1-P}.$$

Moreover, for deterministic prediction from the conservative channel under squared Euclidean loss,

$$\inf_{\widehat{\Psi}} \mathbb{E} \|\Psi - \widehat{\Psi}\|_2^2 = \frac{P}{36}, \quad \inf_{\widehat{\beta}} \mathbb{E} \|\beta - \widehat{\beta}\|_2^2 = \frac{P}{18}.$$

Proof. Each parent has zero mean. The right interface of parent K has equal left and right states σ_K . Hence the conservative integrated flux through every parent interface is

$$\Delta t f(\sigma_K) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4},$$

independent of σ . All conservative flux differences vanish, so the next parent means are identical.

At the same equal interface states,

$$\Psi_K = \Delta t q(\sigma_K) = \frac{1}{2} \cdot \frac{\sigma_K}{3} = \frac{\sigma_K}{6}.$$

Thus $\Psi(\sigma) = \sigma/6$ distinguishes all 2^P sign branches.

Now consider $\beta_K = (\sigma_K - \sigma_{K-1})/6$. If $\beta = 0$, then every neighboring pair is equal, so σ is one of the two constant sign vectors. If $\beta \neq 0$, the cyclic differences determine the entire sign vector: at each edge, a zero difference means the sign is unchanged, while a nonzero difference gives the direction of the sign flip. Hence all nonzero divergence patterns are distinct, and the number of divergence values is $2^P - 1$. Under the uniform branch law,

$$\mathbb{P}(\beta = 0) = \frac{2}{2^P} = 2^{1-P}.$$

For squared-error prediction from the conservative channel, the input channel is constant over all branches. Under the uniform sign law,

$$\mathbb{E}[\Psi] = 0, \quad \mathbb{E}[\beta] = 0.$$

Thus the optimal deterministic predictors are zero. Since

$$\|\Psi\|_2^2 = \sum_{K=0}^{P-1} \frac{\sigma_K^2}{36} = \frac{P}{36},$$

the first error floor follows. For the divergence,

$$\mathbb{E}\|\beta\|_2^2 = \frac{1}{36} \sum_{K=0}^{P-1} \mathbb{E}[(\sigma_K - \sigma_{K-1})^2].$$

Independent neighboring signs differ with probability $1/2$, so

$$\mathbb{E}[(\sigma_K - \sigma_{K-1})^2] = 2,$$

and the second error floor is $P/18$. Finally, Lemma 4.2 verifies the nonnegative cellwise residuals required by Theorem 4.1. $\square$

Table 1 connects the sign family to the cellwise entropy-balance hypothesis. Table 2 summarizes the global information separation. The conservative channel is identical across all branches, but the entropy register and its gauge-invariant divergence are not. The error floors are not distribution-free information-theoretic bounds; they are optimal deterministic squared-error risks for the uniform branch law and the declared entropy targets. The table entries and finite certificate checks are also generated by the accompanying Python scripts using exact rational arithmetic via `fractions.Fraction`. Thus the reported finite identities, residual values, branch counts, and error floors do not depend on floating-point roundoff. The scripts serve as reproducibility certificates for the finite examples; the mathematical claims are proved analytically above.

Table 2

Entropy-accounting sign-family counts and error floors. For each P , all 2^P branches share the conservative channel. The raw entropy register distinguishes all branches, while the gauge-invariant cyclic divergence has $2^P - 1$ values because the two constant sign states share zero divergence. The final columns give the deterministic squared-error floors under the uniform branch law.

P	branches	Ψ values	β values	$\mathbb{P}(\beta = 0)$	$\mathbb{E}\ \Psi\ _2^2$	$\mathbb{E}\ \beta\ _2^2$
2	4	4	3	1/2	1/18	1/9
4	16	16	15	1/8	1/9	2/9
8	256	256	255	1/128	2/9	4/9
16	65,536	65,536	65,535	1/32,768	4/9	8/9

5. Discussion

The results separate one-step information sufficiency from recursive predictive memory. Recursive memory, conditioning, and dissipative decay are studied in the companion paper [7]. Here the focus is narrower: which information channels determine one completed transition and its entropy account?

The SSPRK(2,2) example shows that initial traces are integrator-dependent information objects. Forward Euler uses only the initial trace pair at a parent interface. A multistage method can generate later stage traces from hidden interior values, so the correct ex post object is the realized stage-integrated flux register.

The entropy construction shows that conservative closure and entropy accounting are separate contracts. The conservative channel in Theorem 4.3 determines next parent means but not the augmented entropy account. In learned, reduced, and multiscale finite-volume models [13–16], this suggests that conservation, entropy accounting, stage information, and prediction horizon should be specified separately.

5.1. Limitations and future work

The analysis is intentionally narrow. The SSPRK(2,2) obstruction is a scalar periodic upwind construction, and the entropy-accounting separation uses a particular Burgers–Rusanov entropy-flux convention. The results do not claim a universal lower bound for compressible Euler systems, high-order weighted essentially non-oscillatory reconstructions, nonperiodic grids, or learned closures.

Natural extensions include a general explicit Runge–Kutta theorem characterizing how far interior information can enter stage traces, and a more general entropy-separation theorem identifying flux and entropy pairs that admit conservative collisions with entropy-accounting separation. Another direction is to test the same channel distinctions at adaptive mesh-refinement interfaces and for system examples.

6. Conclusions

A one-step finite-volume closure exists only for declared information that makes the restricted fine update single valued. Complete ordered initial traces close a forward-Euler finite-volume transition with a deterministic two-state numerical flux, but they do not generally close an SSPRK(2,2)/Heun transition. The realized stage-integrated interface flux remains sufficient for the completed transition, but it is an ex post register.

Conservative one-step closure is also not entropy-accounting closure. For a declared Burgers–Rusanov entropy account, the sign family constructed here has identical conservative parent means, conservative interface registers, and next parent means across all branches, while the entropy register and its cyclic divergence vary. Under the uniform branch law, the conservative channel leaves nonzero deterministic squared-error floors for the entropy targets.

The practical message is that coarse, reduced, and learned finite-volume models should specify their retained averages, interface traces, stage registers, entropy-side registers, and prediction horizon before claiming pathwise closure.

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CRediT authorship contribution statement

Antonios Polemitis: Conceptualization, Methodology, Verification, Writing–original draft, Writing–review and editing. Nicholas Christakis: Conceptualization, Methodology, Formal analysis, Software, Verification, Writing–original draft, Writing–review and editing. Dimitris Drikakis: Conceptualization, Methodology, Verification, Writing–review and editing.

Data and code availability

The complete reproducibility package for this study is publicly available in the Certified Simulation repository of the University of Nicosia, <https://github.com/UniversityOfNicosia/certified-simulation>, under `papers/initial-trace-entropy-separation`. The package contains the scripts that generate the computational certificates, CSV files, and LaTeX table fragments reported in the manuscript, together with an independent certificate suite that replays the paper’s exact results in exact rational arithmetic. The table entries and finite certificate checks are also generated by the accompanying Python scripts using exact rational arithmetic via `fractions.Fraction`. Thus the reported finite identities, residual values, branch counts, and error floors do not depend on floating-point roundoff. The scripts serve as reproducibility certificates for the finite examples; the mathematical claims are proved analytically above. No external CFD solver or proprietary software is required.

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